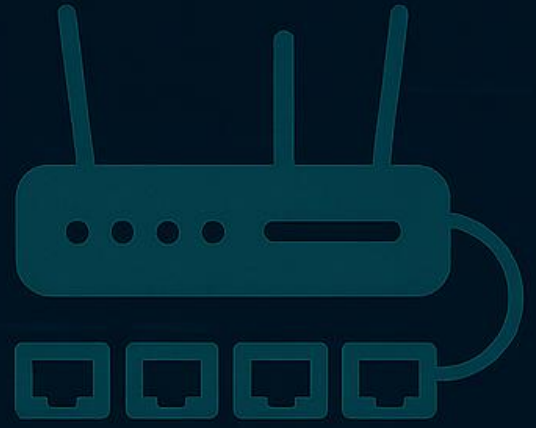




**SECURE YOUR
WAN THE RIGHT WAY**



DMVPN OVER IPSec



Hussain
NETWORKING



Table of Contents

Introduction	3
DMVPN Architecture Overview.....	3
DMVPN Core Technologies	3
DMVPN Phase 3 Behaviour	4
Hub Operation.....	4
Spoke Operation	4
Dual-Hub, Dual-Cloud DMVPN Design	4
Tunnel Source Options (Loopback vs Physical Interface)	5
Physical Interface as Tunnel Source.....	5
Loopback Interface as Tunnel Source	5
Dual Tunnels with Different Sources.....	5
Tunnel Addressing Options	6
Host IPs per Hub/Spoke (Recommended)	6
Using 0.0.0.0/0 (Not Recommended)	6
Hubs Advertise Networks, Spokes Match Networks.....	6
Hub Configurations.....	6
Hub 1 Tunnel.....	6
Hub 1 EIGRP	7
Hub 2 Tunnel.....	7
Hub 2 EIGRP	8
Spoke Configuration Template	8
Spoke Tunnel1 (Primary)	8
Spoke Tunnel2 (Backup)	9
Spoke EIGRP.....	9
IPsec Configuration	10
IKE Phase 1	10
IKE Phase 2	10
IPsec Profile	10
Pre-Shared Key	10
Phase 3 Packet Flow.....	10

DMVPN Phase 3 Over IPsec Technical Explanation & Configuration Guide

Introduction

Dynamic Multipoint VPN (DMVPN) is a Cisco WAN overlay technology that creates scalable, encrypted VPN topologies using multipoint GRE combined with IPsec.

This document explains Phase 3 behaviour, NHRP operations, routing, IPsec encryption, full configurations, and additional tunnel source and addressing options.

DMVPN Architecture Overview

DMVPN Core Technologies

Multipoint GRE (mGRE)

Allows one logical tunnel interface to establish dynamic GRE tunnels to multiple remote peers.

NHRP (Next Hop Resolution Protocol)

Provides tunnel IP to NBMA IP mapping so spokes can discover each other dynamically.

IPsec Transport Mode

Encrypts GRE packets without adding unnecessary tunnel headers, ideal for DMVPN.

DMVPN Phase 3 Behaviour

Hub Operation

Hub configuration includes:

```
ip nhrp map multicast dynamic
```

```
ip nhrp redirect
```

This allows spokes to send the first packet to the hub, then receive an NHRP Redirect to switch to direct spoke-to-spoke forwarding.

Spoke Operation

Each spoke uses:

```
ip nhrp shortcut
```

```
ip nhrp nhs <Hub-Tunnel-IP>
```

These commands enable the spoke to install dynamic shortcuts after receiving NHRP redirects, creating Phase 3 direct tunnels.

Dual-Hub, Dual-Cloud DMVPN Design

Cloud 1 (Primary Path)

```
Tunnel1 → Hub1
```

```
network-id 1
```

```
tunnel key 100
```

```
Lower EIGRP delay (preferred path)
```

Cloud 2 (Backup Path)

```
Tunnel2 → Hub2
```

network-id 2

tunnel key 200

Higher EIGRP delay (used only if Hub1 path fails)

Routing preference is controlled by:

delay 1000 (Primary)

delay 5000 (Backup)

Tunnel Source Options (Loopback vs Physical Interface)

Physical Interface as Tunnel Source

tunnel source Serial1/1

Used for simple ISP WAN links. Tunnel depends directly on the physical link.

Loopback Interface as Tunnel Source

tunnel source Loopback0

Used when multiple WAN uplinks or routing redundancy is required. Loopback must be routable across the WAN.

Dual Tunnels with Different Sources

interface Tunnel1

tunnel source Serial1/1

interface Tunnel2

tunnel source Loopback0

Valid as long as both sources are reachable from the hubs.

Tunnel Addressing Options

Host IPs per Hub/Spoke (Recommended)

Example:

Hub: 126.1.1.1/28

Spokes: 126.1.1.3, 126.1.1.4, 126.1.1.5

Using 0.0.0.0/0 (Not Recommended)

Tunnel interfaces need unique IPs for routing, making this impractical.

Hubs Advertise Networks, Spokes Match Networks

```
network 126.1.1.0 0.0.0.15
```

```
network 126.1.2.0 0.0.0.15
```

Supports scalable overlays.

Hub Configurations

Hub 1 Tunnel

```
interface Tunnel1
```

```
ip address 126.1.1.1 255.255.255.240
```

```
ip nhrp authentication DMVPN
```

```
ip nhrp map multicast dynamic
```

```
ip nhrp network-id 1
```

```
ip nhrp redirect
```

```
tunnel source Serial1/1  
  
tunnel mode gre multipoint  
  
tunnel key 100  
  
tunnel protection ipsec profile DMVPN-PROFILE shared
```

Hub 1 EIGRP

```
router eigrp DMVPN  
  
address-family ipv4 unicast autonomous-system 126  
  
af-interface Tunnel1  
  
no split-horizon  
  
exit-af-interface  
  
network 126.1.1.0 0.0.0.15  
  
network 1.1.1.1 0.0.0.0  
  
exit-address-family
```

Hub 2 Tunnel

```
interface Tunnel2  
  
ip address 126.1.2.1 255.255.255.240  
  
ip nhrp authentication DMVPN  
  
ip nhrp map multicast dynamic  
  
ip nhrp network-id 2  
  
ip nhrp redirect  
  
tunnel source Serial1/1  
  
tunnel mode gre multipoint  
  
tunnel key 200  
  
tunnel protection ipsec profile DMVPN-PROFILE shared
```

Hub 2 EIGRP

```
router eigrp DMVPN  
  
address-family ipv4 unicast autonomous-system 126  
  
af-interface Tunnel2  
  
no split-horizon  
  
exit-af-interface  
  
network 126.1.2.0 0.0.0.15  
  
network 2.2.2.2 0.0.0.0  
  
exit-address-family
```

Spoke Configuration Template

Spoke Tunnel1 (Primary)

```
interface Tunnel1  
  
ip address 126.1.1.X 255.255.255.240  
  
ip nhrp authentication DMVPN  
  
ip nhrp map 126.1.1.1 47.0.0.26  
  
ip nhrp map multicast 47.0.0.26  
  
ip nhrp nhs 126.1.1.1  
  
ip nhrp network-id 1  
  
ip nhrp shortcut  
  
delay 1000  
  
tunnel source Serial1/1  
  
tunnel mode gre multipoint  
  
tunnel key 100  
  
tunnel protection ipsec profile DMVPN-PROFILE shared
```


Spoke Tunnel2 (Backup)

```
interface Tunnel2  
  
ip address 126.1.2.X 255.255.255.240  
  
ip nhrp authentication DMVPN  
  
ip nhrp map 126.1.2.1 47.0.0.22  
  
ip nhrp map multicast 47.0.0.22  
  
ip nhrp nhs 126.1.2.1  
  
ip nhrp network-id 2  
  
ip nhrp shortcut  
  
delay 5000  
  
tunnel source Serial1/1  
  
tunnel mode gre multipoint  
  
tunnel key 200  
  
tunnel protection ipsec profile DMVPN-PROFILE shared
```

Spoke EIGRP

```
router eigrp DMVPN  
  
address-family ipv4 unicast autonomous-system 126  
  
network 126.1.1.0 0.0.0.15  
  
network 126.1.2.0 0.0.0.15  
  
network <Loopback-IP> 0.0.0.0  
  
exit-address-family
```

IPsec Configuration

IKE Phase 1

crypto isakmp policy 10

encr aes 256

hash sha256

authentication pre-share

group 14

lifetime 3600

IKE Phase 2

crypto ipsec transform-set DMVPN-SET esp-aes 256 esp-sha-hmac

mode transport

IPsec Profile

crypto ipsec profile DMVPN-PROFILE

set transform-set DMVPN-SET

Pre-Shared Key

crypto isakmp key DMVPN address 47.0.0.0 255.255.255.192

Phase 3 Packet Flow

1. Spoke registers with hub via NHRP.
2. Routing adjacency forms over GRE/IPsec.
3. Spoke sends first packet → hub receives it.
4. Hub sends NHRP Redirect.
5. Spoke queries NHRP for the destination spoke.
6. Spokes form direct encrypted tunnel.
7. Traffic bypasses the hub.